

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                          | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5<br>HT  | (a) | (i)   |  | All 4 points plotted correctly $\pm <1$ small square division (1)<br>1-3 points plotted correctly $\pm <1$ small square division (0)<br>Straight line of best fit between 1.0 – 6.0 V within $\pm <1$ small square division of each point (1)                                                                                                                                                                            |                 | 2   |     | 2     | 2     | 2    |
|          |     | (ii)  |  | As the voltage increases, the current increases (1)<br>at a decreasing rate (1)<br>Award 2 marks - equal increases of voltage produce smaller increases in current<br><b>Don't accept</b> in a non-proportional way / the increase gets less steep / non-linearly                                                                                                                                                        |                 | 2   |     | 2     | 2     | 2    |
|          | (b) | (i)   |  | Substitution: $\frac{3.0}{0.85}$ (1)<br>= 3.53 [ $\Omega$ ] (1) Accept 3.5 [ $\Omega$ ]<br><b>Don't accept</b> 3.52 [ $\Omega$ ]                                                                                                                                                                                                                                                                                         | 1               | 1   |     | 2     | 2     | 2    |
|          |     | (ii)  |  | 0.85 (1) <b>i.e. denominator from (b)(i)</b><br>+ 0.6 <b>ecf</b> from graph = 1.45 [A] (1)                                                                                                                                                                                                                                                                                                                               |                 | 2   |     | 2     | 2     | 2    |
|          |     | (iii) |  | Substitution and recognition of total current $I$ :<br>$R = \frac{V}{I} = \frac{3.0}{1.45 \text{ ecf}}$ (1)<br>= 2.07 [ $\Omega$ ] (1) Accept 2.1 or 2 [ $\Omega$ ]<br><b>Alternative:</b><br>$R_{\text{wire}} = \frac{V}{I} = \frac{3.0}{0.6 \text{ ecf}} = 5$ (1)<br>then $\frac{1}{3.53 \text{ ecf}} + \frac{1}{5 \text{ ecf}} = 0.483$<br>so $R_{\text{total}} = 2.07$ [ $\Omega$ ] (1) Accept 2.1 or 2 [ $\Omega$ ] |                 | 2   |     | 2     | 2     | 2    |
|          |     | (iv)  |  | By having two loops from the battery there is more current (1)<br>Greater current implies smaller resistance (1)                                                                                                                                                                                                                                                                                                         | 2               |     |     | 2     |       | 2    |